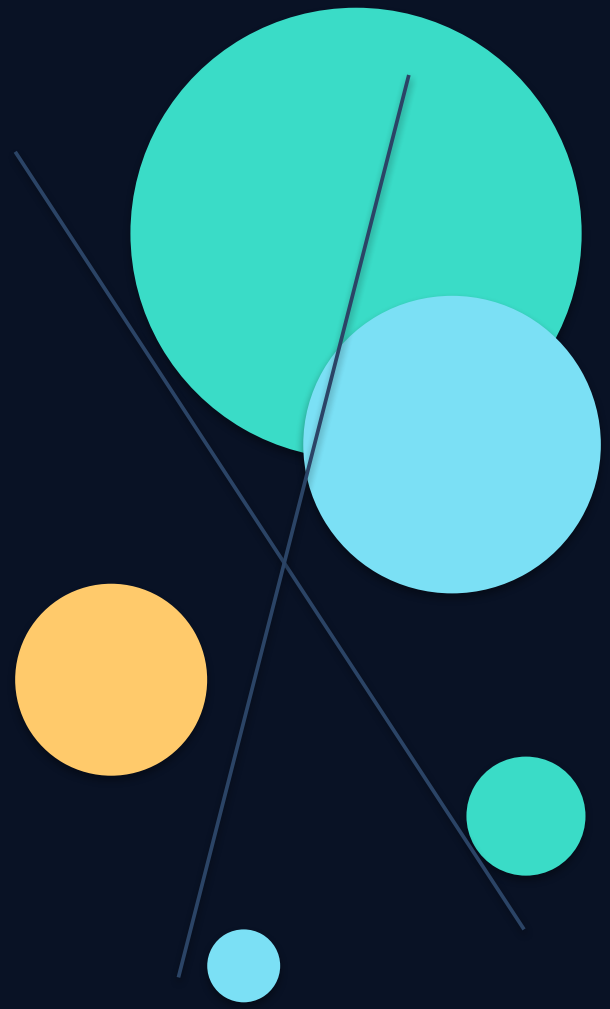
TRACK 2 · QUANTUM OPTIMIZATION

From Portfolios to Quantum Circuits

A practical study of QAOA and quantum machine learning



SUPERPOSITION SQUAD

Rahul Kumar · Alden Antony · Krishna Aggarwal

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One problem, three experiments

We use small, controlled portfolios so we can check our results against a classical answer.

The problem

Choose a set of stocks that balances expected return and risk.

The data

All stock values are synthetic, so the experiments are repeatable.

The rule

A lower objective cost is better, but the portfolio must obey the constraints.

OUR THREE LEVELS

BEGINNER · QAOA

6 stocks → choose 3
Compare all 20 portfolios.

INTERMEDIATE · QML

12 stocks → choose 6
Compare quantum and classical SVMs.

ADVANCED · QAOA

20 stocks → choose 10
Add limits for five sectors.

Our approach: solve classically → test quantum methods → compare the measured outcomes.

Can QAOA find a known solution?

Six stocks; choose three. This gives only 20 possible portfolios, so we can find the exact answer classically.

CLASSICAL ANSWER

Stock 2 · Stock 3 · Stock 6

Best cost: −0.503015

SIMULATOR

4,096 shots
2,783 valid portfolios
67.94% valid rate

IBM HARDWARE

1,024 shots
609 valid portfolios
59.47% valid rate

THE KEY RESULT

The exact best portfolio appeared 29 times in 1,024 hardware shots (2.83%).

A “shot” is one circuit run and measurement. The best observed sample matched the classical answer, but most shots did not return that exact portfolio.

IBM job: db7mlh0r1c5c738kspdg

Why explore quantum machine learning?

Here we ask a different question: can a quantum-state representation help compare candidate portfolios?

1 · Encode

Represent each portfolio as a 12-value input and map it to a quantum state.

2 · Compare

A fidelity score measures how similar two encoded quantum states are.

3 · Classify

An SVM uses the similarity scores to classify selected good/poor examples.

4 · Check

Compare with a standard classical SVM using an RBF kernel.

Initial classification

Quantum-kernel SVM: 100% accuracy
Classical RBF-SVM: 100% accuracy

Ranking comparison

Mean rank correlation: quantum 0.9469; classical 0.9563. The classical model ranked the optimum first in all five recorded tests.

Important: this was synthetic, filtered data. The IBM run was a separate 2-qubit demonstration—not the full 12-qubit QML model.

IBM demo job: db7n3e0r1c5c738ktlbg · 512 shots

More constraints, harder hardware run

20 stocks across five sectors. Select 10 stocks, with no more than two from any sector.

CLASSICAL BASELINE

7,776 feasible portfolios checked
Best cost: −1.111739

SIMULATOR

4,096 shots
10.30% valid
Best cost matched the optimum once.

IBM HARDWARE

256 shots
1 valid portfolio (0.39%)
Exact optimum: 0 hits

COMPILED CIRCUIT **Depth 680** **938 CZ gates**

Deeper circuits may collect more gate errors; noise is a plausible factor, not the only proven cause.

Best valid hardware cost: −0.794773; gap from classical optimum: 0.316966. Pilot parameter search reached its evaluation limit, so convergence was not established.

IBM job: db7n9iuil99s73fogb5g · 256 shots · 20 logical decision qubits

What did we learn?

A useful quantum experiment is more than running a circuit: we must validate and interpret its output.

1 · Check classically

Exact search tells us how good a measured portfolio really is.

2 · Check feasibility

A low cost is useful only if the portfolio follows the rules.

3 · Check hardware

Shots, depth, and noise affect what we observe.

We demonstrated QAOA, quantum-kernel learning, simulation, and real IBM hardware runs.
We did not demonstrate quantum advantage.

Next: improve parameter search, test constraint-aware circuits, and run repeated, controlled comparisons.

Thank you · Questions?

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